

UZ2400

U-Power500 Module User Manual

Application Note

AN-2400-13

The content of this technical document is subject to change without notice. Please contact UBEC for further information.

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1. General Information

The U-Power500 module is an IEEE 802.15.4 ZigBee™ compliant solution that satisfies the requirements of low-power and long-range wireless sensor applications. It integrates UBEC's UZ2400 2.4GHz transceiver with UBEC's power amplifier UP2268 to extend the transmitting and the receiving distance at low current consumption. Both the low current consumption and long distance capability are especially suitable for battery-limited applications. The module's flexible and reliable interface allows the users to develop configurations appropriate for their applications.

The U-Power500 module, derived from UBEC's U-Force module, operates within the ISM 2.4 – 2.5 GHz frequency band and contains UBEC's UZ2400, UP2268 and other necessary components such as crystal oscillator, inductors and capacitors. The 802.15.4 compliant transceiver within the UZ2400 chip features a maximum of 5 MHz serial interface SPI bus for control and data transfer. It is a simple SPI interface slave device consisting of a 4-wire bus (SCLK, SI, SO, and SEN) that provide accesses to various subunits such as MAC/BB/RF control/status registers, TXFIFOs, RXFIFOs and security key table.

The UP2268 is an RF front-end IC integrating a high-efficiency, linear power amplifier (PA), a switch and a low noise amplifier (LNA). Its low current consumption and inherent power gain enable the designers to meet the challenges of the battery operation and long distance requirements demanded by many applications. In addition, the small module form factor (QFN3x3) saves the valuable board space.

2. Features

- ☐ 2.4GHz IEEE 802.15.4 and ZigBee™ Compliant
- ☐ 3.0~3.6V Operation
- ☐ Sleep Current: 3uA (typical)
- ☐ >500m Range (Environment dependent)
- ☐ PCB Antenna
- ☐ 2 GSG (ground-signal-ground) interfaces provided
- ☐ Receiver Sensitivity: -101dBm
- ☐ Tx Power : 10dBm
- ☐ Tx Current : 54Ma
- ☐ Rx Current : 26mA
- ☐ Size: 48.3mmx 14mm

* ZigBee™ is a trade mark of ZigBee Alliance.

3. PCB of U-Power500 Module

Two types of the U-Power500 module are available, depending on the antenna configurations.

Figure 1 shows the top view of a U-Power500 Printed Circuit Board (PCB) with a printed circuit antenna. The PCB size : is 48.3mm x 14 mm.

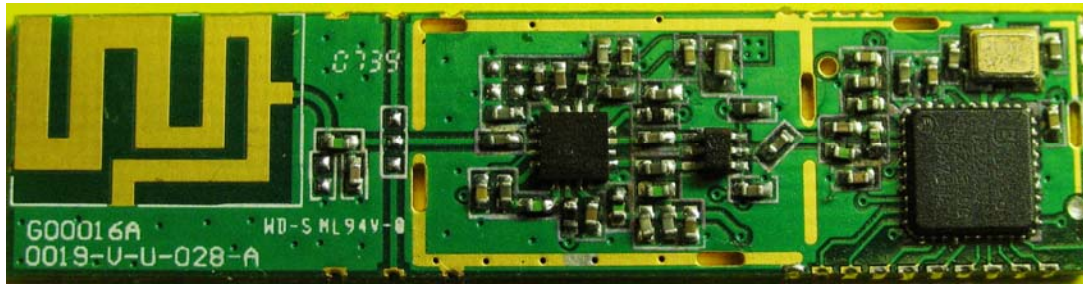


Figure 1. UZ2400 U-Power500 Printed Antenna Module

4. U-Power500 Pin Configuration

UZ2400 U-Power500 module pin assignments and their functions are described below.

Pin Number	Pin Name	Type	Description
1	GPIO0	DIO	General purpose digital I/O, also used as an external PA enable
2	GPIO1	DIO	General purpose digital I/O, also used as an external TX/RX switch control
3	GPIO5	DIO	General purpose digital I/O
4	GPIO4	DIO	General purpose digital I/O
5	GPIO2	DIO	General purpose digital I/O, also used as an external TX/RX switch control
6	GPIO3	DIO	General purpose digital I/O
7	RESETN	DI	Global hardware reset pin, active low
8	WAKE	DI	External wake up trigger, active high / low can be programmable.
9	INT	DO	Interrupt pin to micro-processor : Level trigger, Hi / Low programmable
10	SO	DIO	Serial interface data output from UZ2400 or I2C clock
11	SI	DIO	Serial interface data input to UZ2400 or I2C data in/out
12	SCLK	DI	Serial interface clock
13	SEN	DI	Serial interface enable
14	CLKOUT	DIO	20/10/5/2.5 MHz Clock output
15	XTAL32N	AI	32 kHz Crystal input (-) for internal RTC used

16	XTAL32P	AI	32 kHz Crystal input (+) for internal RTC used
17	GND	Ground	Ground
18	VCC	Power	3V
19	VCC	Power	3V
20	GND	Ground	Ground

UZ2400 U-Power500 module pin names and pin number allocation map are shown in Figure 2.

It can interface with other devices such as a sensor, an LED, a host controller, a push button, a joystick or power relays through GPIO signals.

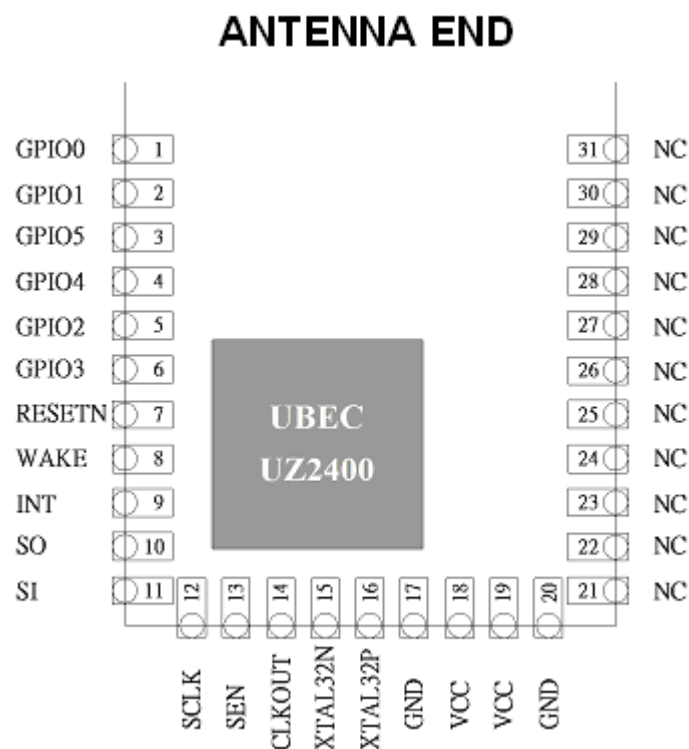


Figure 1. U-Power500 Module Pin Names and Location Map

5. Electrical Specifications

ITEM	SPECIFICATION	UNIT
Frequency range	ISM band 2.4 – 2.5	GHz
Input voltage	3	V
Current consumption	TX 54.0 RX 26.0	mA
TX Output power	10 (Typ)	dBm
TX EVM	10% ~18% (2405 ~ 2480 MHz)	%
RX sensitivity	-100	dBm

For detailed functional descriptions and electrical characteristics of the UZ2400 chip, please refer to UZ2400 datasheet.

5.1. U-Power500 RF module TX output power

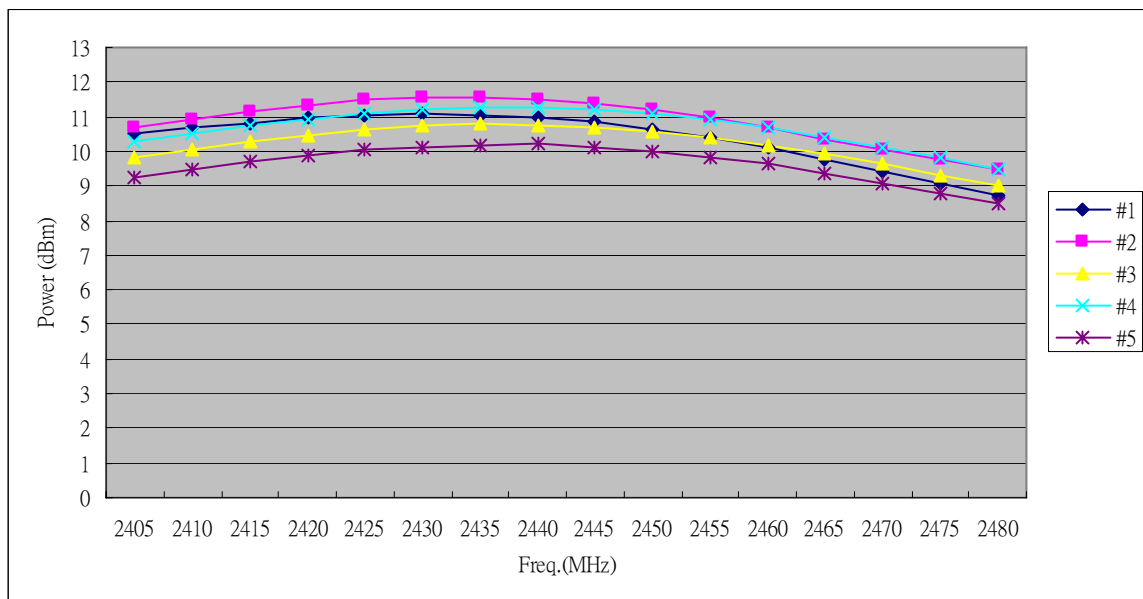


Figure 2. Typical U-Power500 RF Module TX Output Power Test Results

5.2. TX EVM

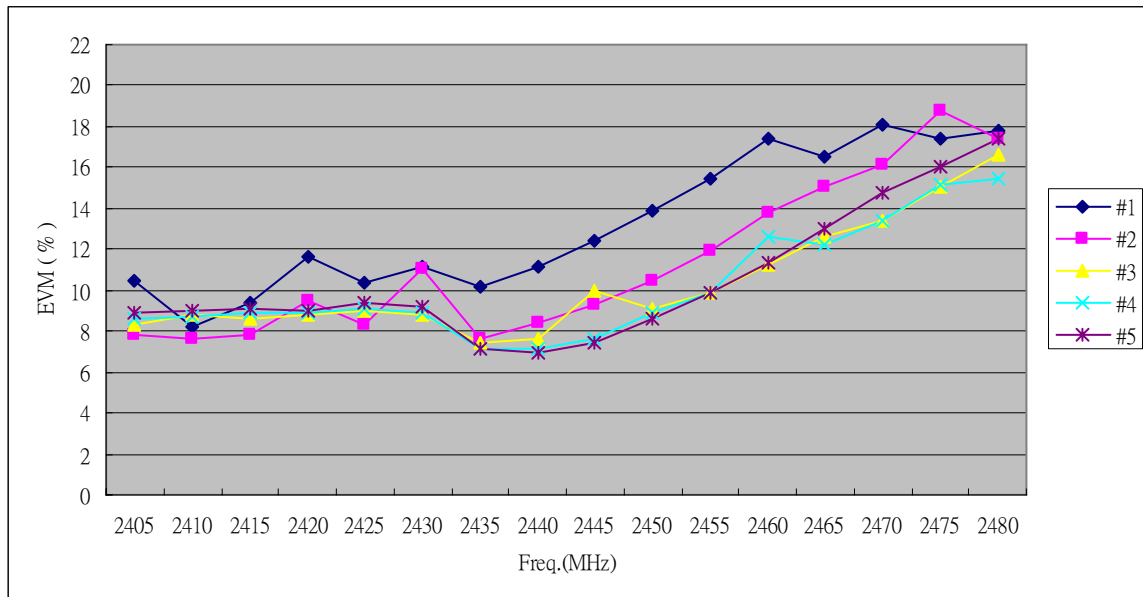


Figure 3. Typical U-Power500 RF Module TX EVM Test Results

5.3. RX sensitivity

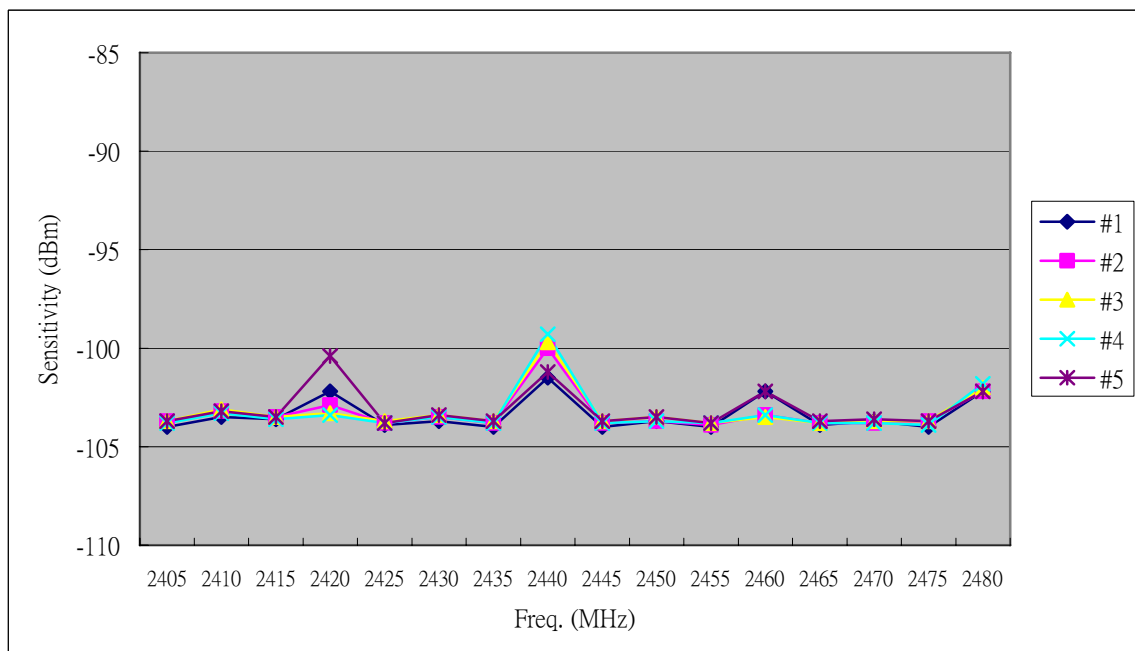


Figure 4. Typical U-Power500 RF Module RX Sensitivity Test Results

5.4. LOS (Line-of-Sight) Results

Software: UBEC developed program:19TM06

Platform: Window XP/ASUS Notebook

Module: U-Power500

Criteria: PER (Packet-Error-Rate) <5%

- ☐ [Google Real Estate Search](#)
- ☐ [Photos from Panoramio](#)
- ☐ [Gas Prices from GasBuddy.com](#)
- ☒ [Distance Measurement Tool](#)
- ☐ [Our Earth as Art](#)
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Distance Measurement Tool

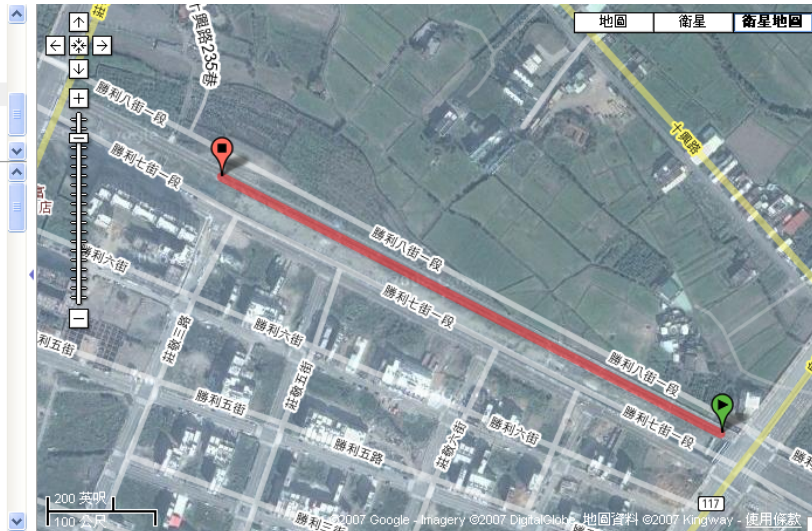
Click on the map to trace a path you want to measure.

Units: ☒ Metric ☐ English

Total distance: 520.84 m

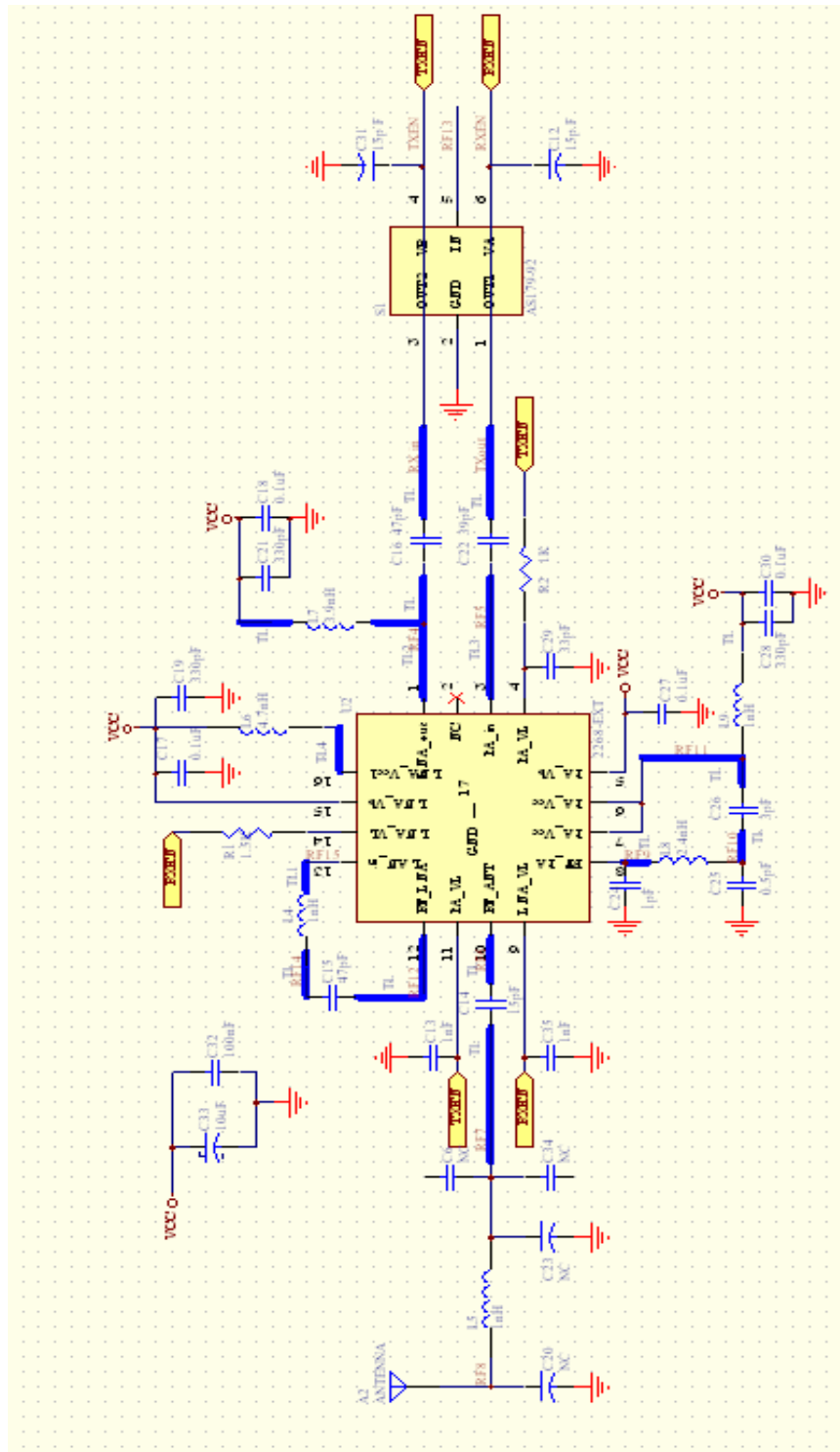
Delete last point

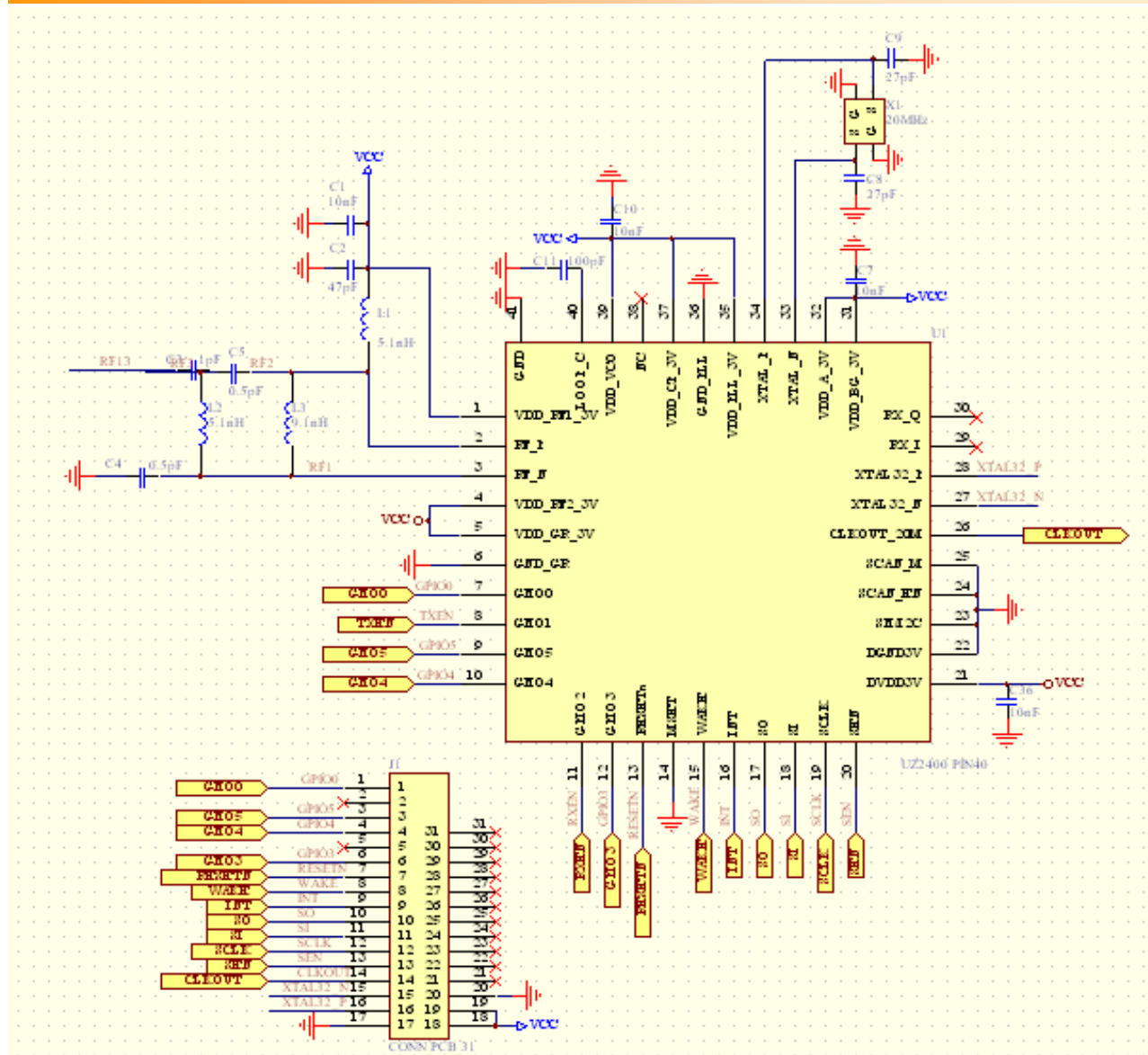
Reset



6. Schematic

The schematic below is suitable for operation with a bias voltage between 3.0 and 3.6V. If a lower bias voltage between 2.7 and 3.0V is desired, certain resistors in this schematic have to be replaced. Please consult UBEC for further information.





7. Mechanical Dimension Drawing

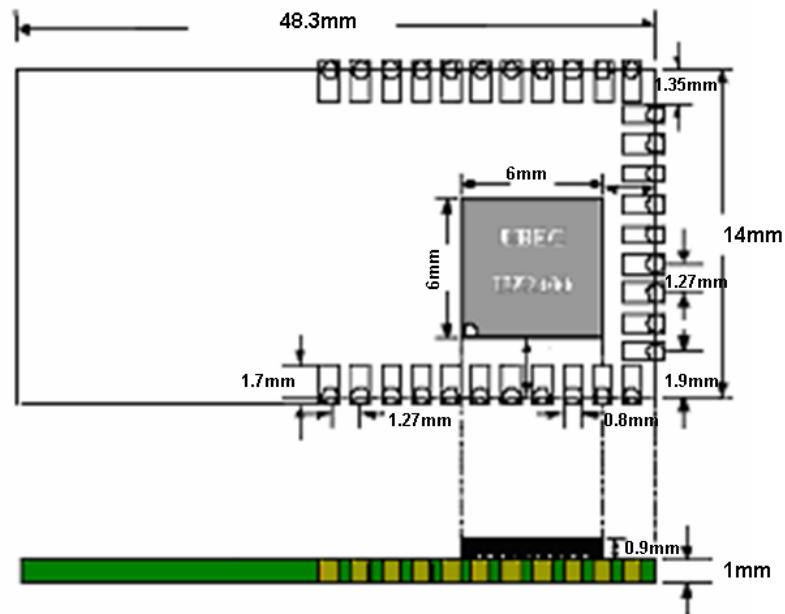


Figure 5. Printed Antenna Module Dimensions

8. U-Power500 Antenna Topology

Notice:

- (1) Antenna should be put on the edge of the system.
- (2) Be sure that there is no obstacles (components and ground) in the radiation direction
- (3) Do not put any ground plane and/or circuit beneath the antenna region for the system boards.

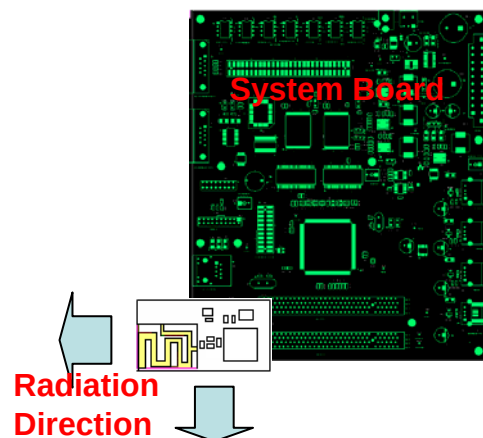


Figure 7. Recommended Antenna Location

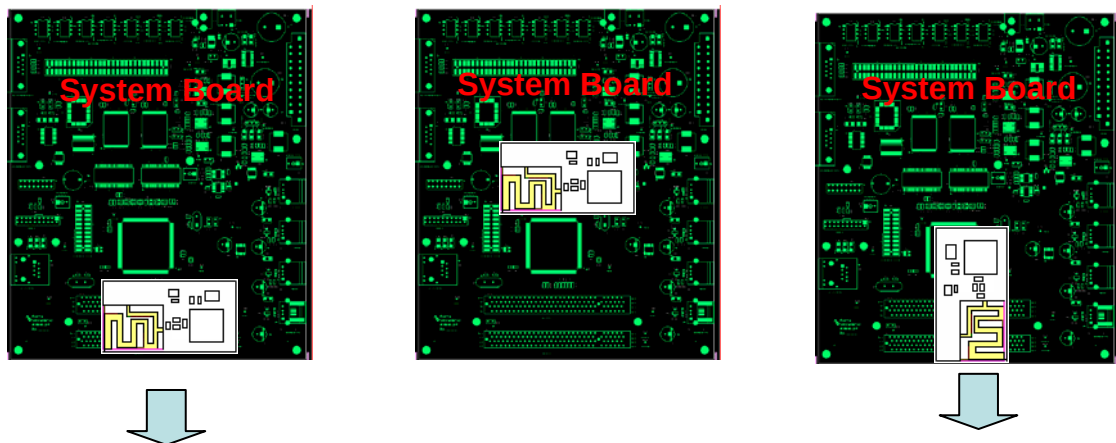
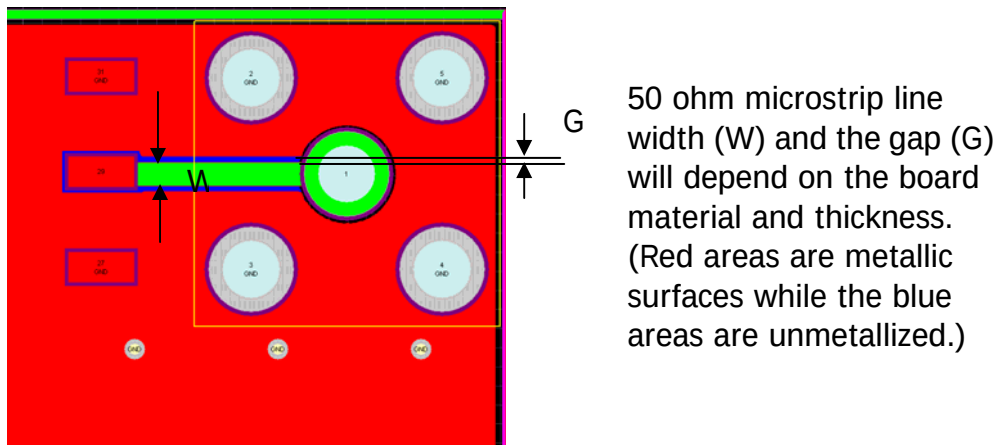


Figure 8. Example of Poor Antenna Locations

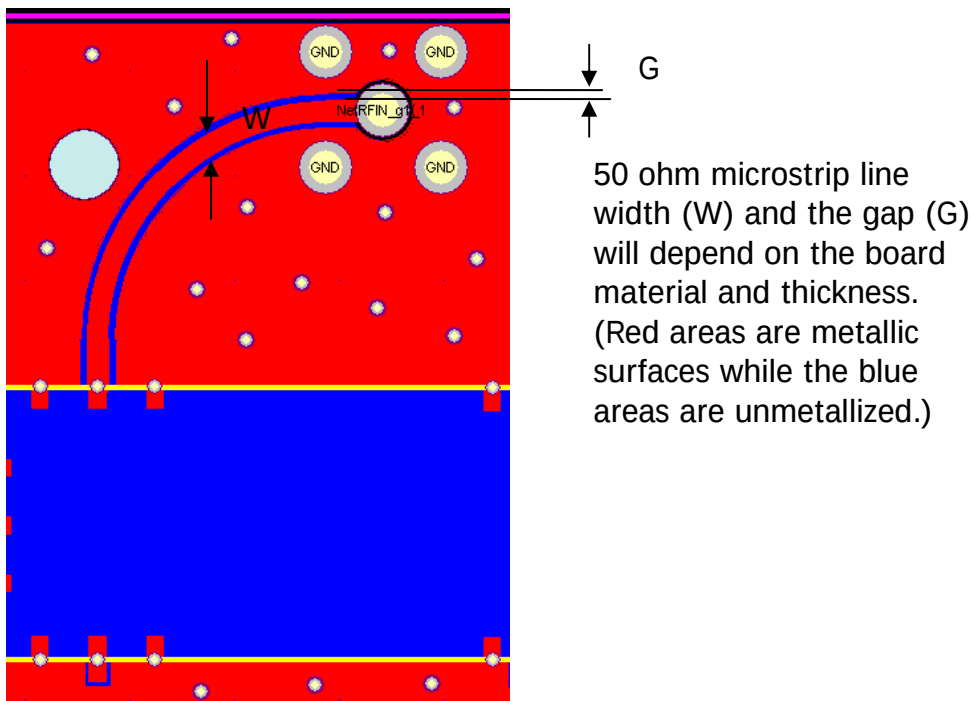
9. Ground-Signal-Ground (GSG) Interface Design

To prevent a huge RF power loss from the GSG to the system board, the recommended rules for laying the micro-strip lines should be obeyed. Please refer to UBEC's Application Note: AN-PCB1-01.

9.1. For 4-layer PCB

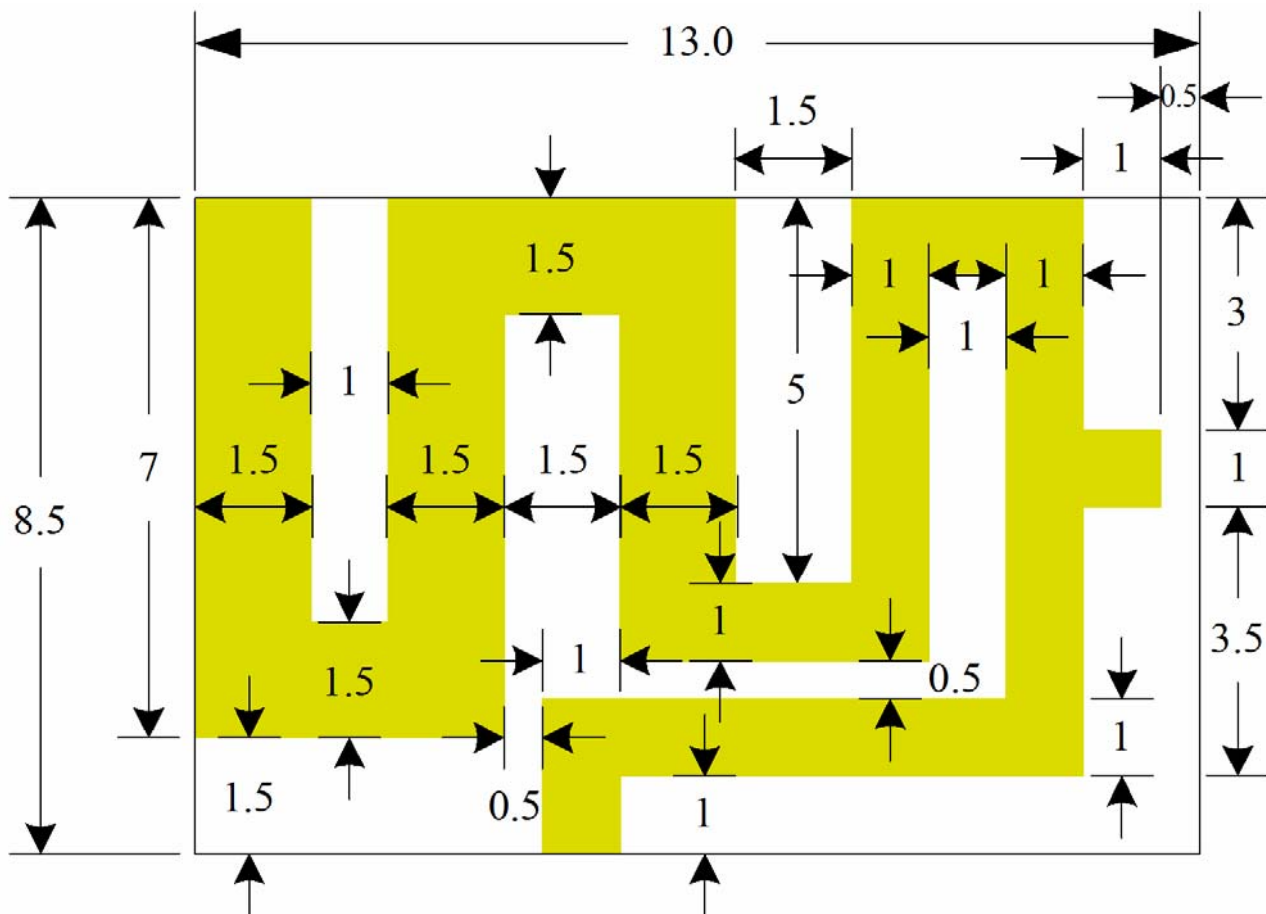


9.2. For 2-layer PCB



10. Appendix

10.1. Meander antenna pattern and its dimensions



Unit : mm

Figure 9. U-Power500 Meander Antenna Pattern and Dimensions

Revision History

Revision	Date	Description of Change
0.0	2007/10/30	Initial release

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